

Neural network control of chaos in a class of impact vibration systems

Ningzhou Li¹, Xinyu Bai¹ , Xiaojuan Wei¹, Xiaoqi Li¹ and Qingli Zhang¹

Abstract

This study examines the challenge of regulating chaotic motion within a specific category of single-degree-of-freedom impact vibration systems. By utilizing principles of mechanics and system dynamics theory, the authors developed a mathematical model for the system's dynamics. To facilitate chaos control, a double-hidden-layer radial basis function neural network (DRBFNN) chaos controller, characterized by its robust nonlinear properties, is proposed. This controller demonstrates enhanced nonlinear fitting capabilities, thereby effectively accommodating the system's pronounced nonlinear and non-smooth characteristics. Furthermore, the Harris Hawks Optimization (HHO) algorithm is employed to optimize the parameters of the controller. The chaotic behavior of the system is investigated through bifurcation diagrams and Lyapunov exponent spectra. A fitness function is formulated based on the distances between adjacent points on the Poincaré section, which guides the parameter optimization process of the HHO algorithm, thereby validating the efficacy of the control strategy and ensuring the efficiency and precision of the parameter search for the controller. Simulation results indicate that the HHO-DRBFNN control approach is successful in transforming chaotic motion into periodic motion, exhibiting commendable dynamic response performance.

Keywords

impact vibration system, chaos control, Harris Hawks optimization (HHO), double-hidden-layer radial basis function neural network (DRBFNN), data driven control

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Introduction

In engineering practice, many vibration-impact phenomena are caused by clearances. For example, the clearance between wheels and rails in a moving train can cause impact vibrations.¹ This repeated impact behavior can lead to bifurcation of system motion and eventually trigger chaotic motion. One of the main causes of structural fatigue damage and even complete failure of some mechanical system components is chaotic motion. To avoid such damage to mechanical structures, scholars have conducted relevant research on chaos control in impact vibration systems. The rapid development of intelligent optimization algorithms in recent years has provided scholars with new ideas for controlling chaotic motion.

de Souza et al.^{2,3} investigated the damping control technique for chaotic dynamics within a specific category of single-degree-of-freedom vibro-impact systems,⁴ successfully employing the OGY method to regulate their chaotic dynamics. In their examination of single-degree-of-freedom vibration systems, Shaw and Holmes⁵ identified a range of complex dynamic phenomena, including fundamental periodic vibrations,

subharmonic impact vibrations, and chaotic behavior. They emphasized that the unique characteristics of grazing bifurcations render traditional research methodologies for smooth systems ineffective in the context of bifurcation analysis for non-smooth systems. Nordmark,⁶ in his study of a vibration system characterized by a single degree of freedom that is constrained rigidly and subjected to sinusoidal excitation, discovered that the modification of a single parameter could facilitate the transition from stable periodic motion to aperiodic motion through grazing bifurcations. He proposed a criterion for identifying grazing bifurcations in the system based on qualitative analysis of singular points. Liu et al.⁷ introduced an innovative control approach utilizing fuzzy sliding mode control (FSMC)

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to manage the chaotic vibrations of an axial string. Huang et al.⁸ developed a particle swarm optimization (PSO)-enhanced sliding mode control strategy aimed at mitigating chaos in permanent magnet linear motors. Li and Cui⁹ constructed a six-dimensional hyper-chaotic particle motion system, analyzed its characteristics, and successfully attained regulation of its erratic dynamics. Wei et al.¹⁰ introduced a collaborative control approach characterized by two parameters employing LRS-QPSO-optimized support vector machines (SVM) for chaos control in a single-degree-of-freedom vibro-impact system characterized by soft constraints. Wang et al.¹¹ suggested an external periodic force control strategy to address chaotic phenomena in single-degree-of-freedom impact vibration systems, effectively managing the system's chaotic behavior through the design of an external periodic force feedback controller. Barış et al.¹² proposed the REMD-LSTM-TAVO model, which combines REMD decomposition, LSTM network, and TAVO optimization to improve wind speed prediction accuracy, outperforming multiple comparison models.

In the field of chaos control, radial basis function neural networks (RBFNNs) have become a key bridge connecting intelligent algorithms and chaos control due to their unique nonlinear mapping capabilities and data-driven characteristics, which enable effective capture of chaotic dynamic features without relying on precise mathematical models of the system. Li et al.¹³ proposed a GWO-RBFNN two-parameter collaborative control method for chaos control in permanent magnet synchronous motors. Li et al.¹⁴ developed a chaos control method for impact vibration systems based on RBFNN and gradient descent method, with effective simulation verification. Liang¹⁵ analyzed bifurcations and chaos in impact vibration systems and designed an RBFNN controller combined with the improved particle swarm optimization (IPSO) algorithm to achieve effective control. Zhang et al.¹⁶ proposed an RBFNN-PSO synchronization method to optimize controller parameters of the Sprott B chaotic system for anti-noise performance, realizing master-slave synchronization and applying it to image encryption, with good encryption effects verified by simulations.

The Harris Hawks Optimization (HHO) algorithm was introduced by Heidari et al.¹⁷ and Sihwail et al.¹⁸ This algorithm emulates the hunting behavior of Harris hawks and encompasses three distinct phases: exploration, transition, and exploitation, thereby providing effective solutions to a range of optimization challenges. Subsequently, Ewees and Elaziz¹⁹ developed the CMVHHO algorithm, which incorporates chaos theory into the HHO framework to optimize the parameters of Multi-Verse Optimizer (MVO) and enhance local search capabilities. Hu and Xiong²⁰ introduced chaotic search into Harris hawks to enhance exploitation capabilities. Altan²¹ compared PSO and HHO algorithms in optimizing UAV attitude and altitude controller

parameters, showing that HHO-optimized PID performs better in multi-path tests. Ji and Wang²² proposed an improved HHO algorithm with chaotic mapping and adaptive weights to optimize 3D distribution paths of UAV rescue materials. Liu et al.²³ proposed an ANN-HHO hybrid model to optimize ANN parameters, achieving optimal accuracy, efficiency, and convenience in data prediction. However, research on introducing the HHO algorithm into chaotic impact vibration systems remains scarce.

Vibro-impact systems are strongly nonlinear systems with non-smooth characteristics, so control methods applicable to smooth systems cannot be directly applied. In engineering practice, the service environment of vibro-impact systems is complex, making accurate modeling difficult. Among relevant control methods, Sliding Mode Control (SMC) and Linear Quadratic Regulator (LQR) control rely on the system model and are unsuitable for data-driven chaotic control of vibro-impact systems. Proportional-Integral-Derivative (PID) and fuzzy control are model-independent, but PID is more suitable for linear systems, while fuzzy control involves complex fuzzy rule formulation that depends on designers' experience.

In contrast, The HHO-DRBFNN approach operates independently of the precise model of the controlled system and does not depend on the subjective expertise of designers. Building upon this foundation, the present study introduces a DRBFNN controller optimized via the HHO algorithm for the chaotic control of a specific category of single-degree-of-freedom vibro-impact systems, improving the system's dynamic performance. The research approach is: first, establish the dynamic model and analyze chaotic characteristics; second, design a dual-hidden-layer radial basis function neural network controller and optimize its parameters using HHO; finally, verify the control effect of HHO-DRBFNN by comparing with HHO-RBFNN and TSA-DRBFNN methods.

The contributions of this study are delineated as follows: The DRBFNN chaotic controller is designed, which has fewer parameters and stronger nonlinear fitting ability compared with traditional controllers, better adapting to the strong nonlinearity and non-smoothness of vibro-impact systems. The HHO algorithm is introduced to optimize controller parameters, improving the efficiency and accuracy of parameter search.

The organization of the paper is as follows: Section 1 outlines the dynamic model of a chaotic impact vibration system characterized by a single degree of freedom; Section 2 details the proposal of the DRBFNN controller; Section 3 discusses the parameter optimization of the controller through the HHO algorithm; Section 4 provides a simulation analysis and comparative evaluation; and finally, Section 5 provides a summary of the study's findings and delineates prospective directions for subsequent research.

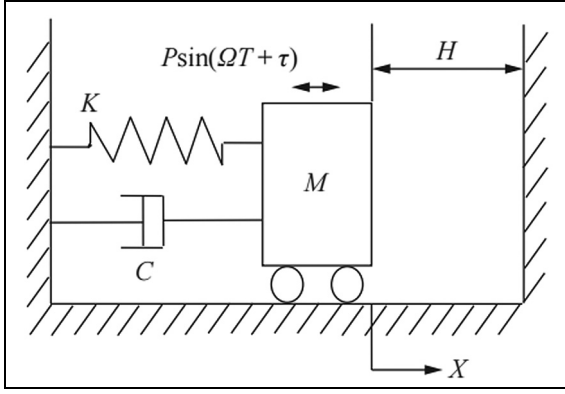


Figure 1. The mechanical model.

An examination of chaotic behavior in a single-degree-of-freedom impact vibration system

Figure 1 illustrates the mechanical model examined in this study, denoted as M , is situated on a frictionless plane and is connected to a linear spring characterized by a stiffness constant K on the left side, and a linear damper characterized by a damping coefficient C is positioned on the right side.²⁴ The mass block M undergoes vibrations induced by a harmonic excitation force represented by the expression $P \sin(\Omega T + \tau)$. In this equation, X denotes the displacement of the mass block. The distance from the mass block M to the right rigid constraint at the equilibrium position is designated as H .²⁵

The differential equation of motion is as follows if the collision duration is disregarded²⁶:

$$\begin{cases} M\ddot{X} + C\dot{X} + KX = P \sin(\Omega T + \tau), X < H \\ \dot{X}_+ = -R\dot{X}_-, X = H \end{cases} \quad (1)$$

From the above equation, the following equation can be obtained without loss of generality through dimensionless transformation:

$$\begin{cases} \ddot{x} + 2\xi\dot{x} + x = \sin(\omega t + \tau), x < h \\ \dot{x}_+ = -R\dot{x}_-, x = h \end{cases} \quad (2)$$

Where $\ddot{X}$, $\dot{X}$, and X represent the acceleration, velocity, and displacement of the mass block M ,²¹ respectively. $\dot{x}_+$ and $\dot{x}_-$ are the instantaneous velocities of the mass block before and after collision, respectively. ξ is the damping. ω is the harmonic excitation frequency. τ is the phase. H is the clearance.

To reveal the dynamic evolution mechanism of the system in Figure 1, the instantaneous section after collision is selected, where $\sigma = \{(x, \dot{x}, \theta) \in R^2 \times S^1 | x = h, \dot{x} = \dot{x}_+\} = \omega t \bmod(2\pi)$ is chosen as the Poincaré section. This study endeavors to model the interplay between the chaotic dynamics of the system. And the

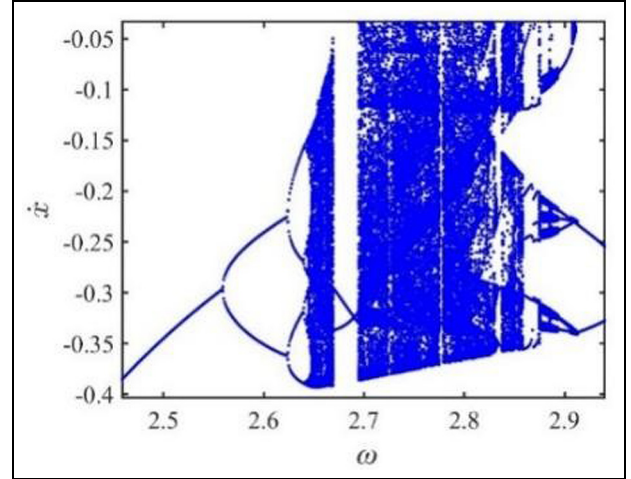


Figure 2. The bifurcation diagram of system as ω changes.

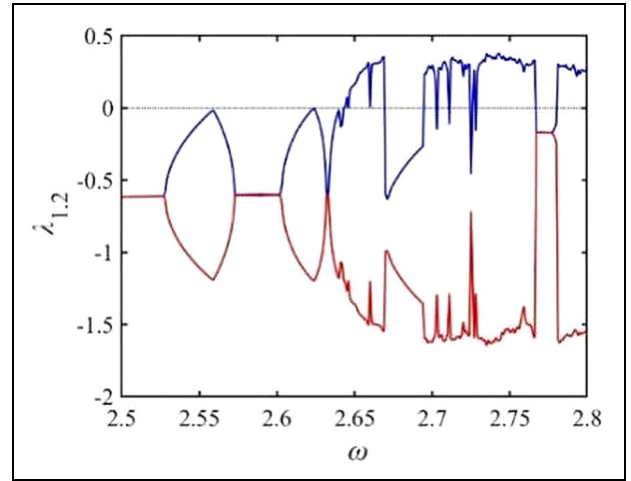


Figure 3. The Lyapunov spectra of system.

change in excitation frequency under the parameters $\mu = 0.2$, $R = 0.8$, and $h = 0.05$. The bifurcation diagram and Lyapunov spectrum of the system are shown in Figures 2 and 3.⁴

As shown in Figure 2, when the excitation frequency passes through 2.56 and 2.62, the system motion evolves from stable 1-1 periodic motion (to identify the type of system motion, the symbol $n\omega$ - p is used, where n is the number of excitation periods and p is the number of collisions) to 2-2 periodic motion, then to 4-4 periodic motion, and then chaotic motion appears. However, when it continues to increase, the chaotic motion becomes periodic motion.

Figure 3 is the Lyapunov exponent spectrum. As a quantitative index, the Lyapunov exponent serves as a significant metric for assessing the stability and chaotic properties of dynamic systems. λ_1 is the maximum Lyapunov exponent. When $\lambda_1 < 0$, the system has stable n - p periodic motion. When $\lambda_1 > 0$, chaotic motion occurs. When $\lambda_2 = 0$ and $\lambda_2 < 0$, period-doubling bifurcation occurs.

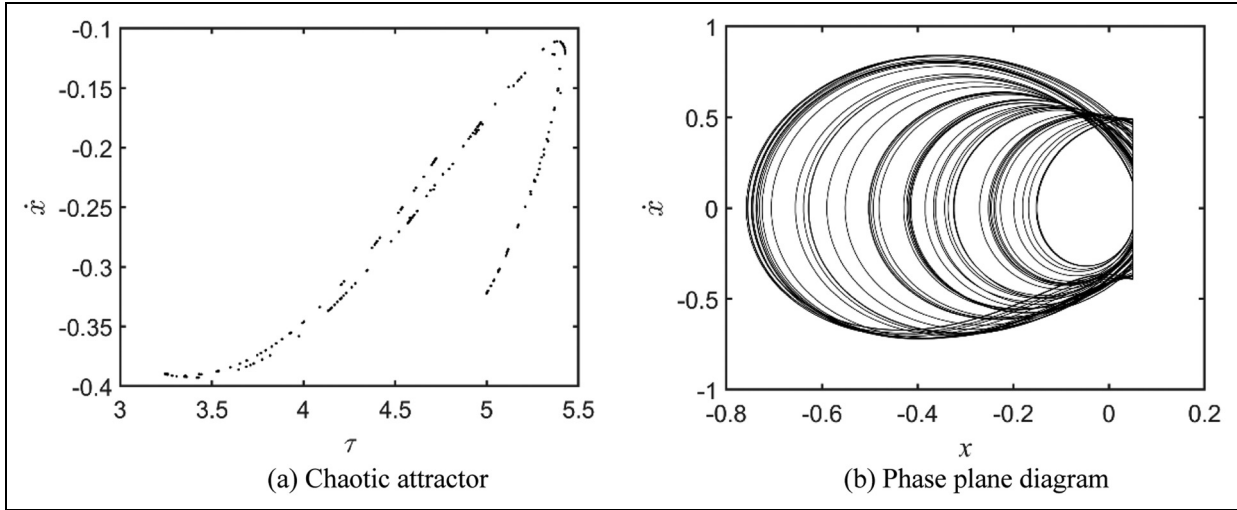


Figure 4. Chaotic attractor and phase plane diagram ($\omega = 2.655$): (a) chaotic attractor and (b) phase plane diagram.

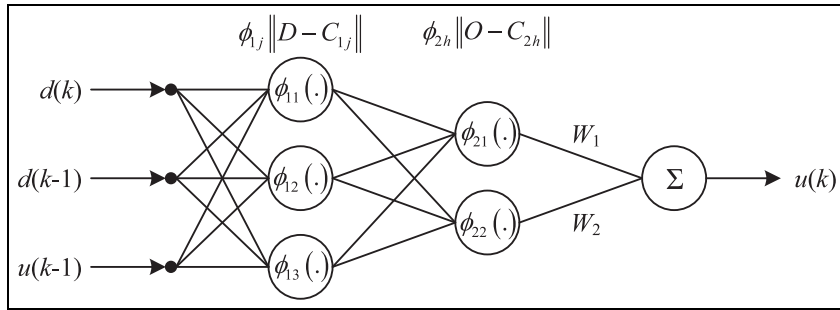


Figure 5. Structure of the controller.

When the parameters are $\omega = 2.655$, $\xi = 0.2$, $R = 0.8$, $h = 0.05$, the chaotic attractor and the phase plane diagram of the Poincaré section are depicted in Figure 4. Figure 4(a) reveals that the Poincaré section manifests as a point set characterized by a hierarchical structure. In contrast, the phase plane illustrated in Figure 4(b) is non-repetitive and disordered, signifying that the system is engaged in chaotic motion.

DRBFNN-based chaotic controller

Controller structure

This paper designs a controller based on DRBFNN,²⁷ which includes two hidden layers: the first layer is responsible for initially extracting basic nonlinear features, and the second layer further integrates and maps these features.

Figure 5 illustrates the architecture of the controller. As depicted, the architecture consists of the input layer, followed by the first hidden layer, the second hidden layer, and concluding with the output layer.²⁸ The three inputs of the controller are the distances between adjacent projected points on the Poincaré section after two consecutive iterations, reflecting the system's trend toward stable periodic motion (i.e. $d(k) = \|X(k) - X(k-1)\|$,

$d(k-1) = \|X(k-1) - X(k-2)\|$, $u(k-1) = \|d(k) - d(k-1)\|$, $X(k)$ is the state variable value of the controlled system at the k th iteration of $X = [\dot{x}, \tau]$, and $u(k-1)$ is the adjustment amount of ω .

The initial hidden layer comprises three nodes, while the subsequent hidden layer consists of two nodes, the radial basis function neural network has five hidden layer nodes, and all activation functions are Gaussian functions (i.e. $\phi_{sk}(D - C_{sk}) = \exp\left(-\frac{\|D - C_{sk}\|^2}{2\sigma_{sk}^2}\right)$, C_{sk} serves as the central point within the hidden layer, and σ_{sk} is the width of the radial basis function), and only the second hidden layer is connected to the output layer with weights W_1 and W_2 . The output of the controller is $u(k) = \sum_{h=1}^2 w_h \phi_{2h}(O, C_{2h}) = \Delta\omega$ ($O = [\phi_{11}(D, C_{11}), \phi_{12}(D, C_{12})]^T$).

Controller advantages

In the field of chaotic control, the controller's ability to fit system nonlinearity and non-smoothness directly determines the control effect. The DRBFNN-based controller designed in this paper (structure shown in Figure 5) has the following advantages:

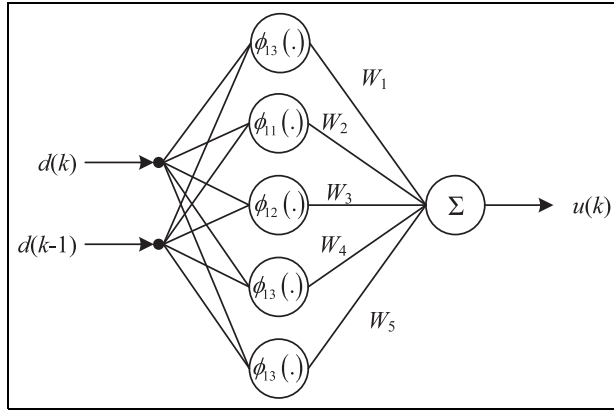


Figure 6. Structure of the RBFNN controller.

- Stronger nonlinear mapping ability:** The proposed controller adopts a dual-hidden-layer structure. Through multi-level mapping, it improves the fitting accuracy for the strong nonlinearity and non-smoothness of the vibro-impact system, making it more adaptable to the complex dynamic behavior of the controlled system. In contrast, the RBFNN-based controller (structure shown in Figure 6) with a single hidden layer has poor nonlinear mapping ability when dealing with strongly nonlinear and non-smooth systems.
- Fewer parameters to be optimized:** The proposed controller reduces parameter redundancy through hierarchical feature extraction, resulting in fewer hidden layer nodes and correspondingly fewer parameters to be optimized. In contrast, the single-hidden-layer RBFNN-based controller requires more hidden layer nodes to achieve better fitting performance, leading to more parameters to be optimized.
- Stronger learning ability:** Compared with the controller shown in Figure 6, the proposed controller incorporates not only $d(k)$ and $d(k-1)$ but also $u(k)$ as input variables. Thus, it can better learn the variation laws of the dynamic characteristics of the controlled system contained in the input-output data, thereby ensuring better control effects.

Parameter optimization of DRBFNN controller based on HHO

HHO algorithm

The HHO algorithm represents a meta-heuristic methodology developed to tackle optimization problems by emulating the hunting strategies exhibited by Harris hawks in their natural environment. The hunting strategy of Harris hawks mainly includes three stages: exploration, conversion, and exploitation, which are mathematically modeled to solve target problems. The exploration stage mainly searches for potential prey locations through random search and global

exploration, while the exploitation stage optimizes solution quality through local search and precise capture. Based on this, the hunting behavior of the HHO algorithm can be mathematically simulated through the following three steps:

Exploratory phase. In the exploration phase, Harris hawks conduct a stochastic search and engage in global exploration to locate potential prey. The methodology for updating their position is articulated as follows:

$$X(t+1) = \begin{cases} X_{rand}(t) - r_1 |X_{rand}(t) - 2r_2 X(t)|, & q \geq 0.5 \\ [X_{rabbit}(t) - X_m(t)] - r_3 [r_4(ub - lb)], & q < 0.5 \end{cases} \quad (3)$$

Where $X_i(t)$ and $X(t+1)$ are the current and next generation individual positions, respectively; t is the iteration number; $X_{rand}(t)$ is the position of the randomly selected individual; $X_{rabbit}(t)$ is the prey position (individual position with the best fitness); lb and ub are the upper and lower bounds of the search region; r_1, r_2, r_3, r_4 , and q represents a set of random numbers that are confined within the interval (0,1); $X_m(t)$ is the mean status of individuals within the population; the expression is

$$X_m(t) = \sum_{k=1}^M X_k(t) / M \quad (4)$$

where $X_k(t)$ represents the position of the k -th individual within the population; while M denotes the total size of the population.

Conversion between search and hunting stage. The HHO algorithm employs a dynamic approach that alternates between search and hunting behaviors, which are influenced by the absolute magnitude of the prey's escape energy during the course of the escape process. The energy required for escape, denoted as $|E|$, determines the phase of the process. Specifically, when the absolute value of the escape energy is greater than or equal to 1, the system transitions into the search stage.²³ Conversely, when the absolute value of the escape energy is less than 1, the system enters the hunting stage. The escape energy is characterized as follows:

$$E = 2E_0 \left(1 - \frac{t}{T}\right) \quad (5)$$

Where E_0 is the initial energy possessed by the prey, with a value range of $[-1, 1]$; t is the current iteration number; T is the maximum iteration number.

Hunting stage. In the HHO algorithm, the strategy for updating positions during the hunting phase is characterized by a piecewise function that incorporates the magnitude of the escape energy, denoted as $|E|$,

alongside a random variable r . The specifics of this approach are outlined as follows:

1. When $r \geq 0.5$ and $0.5 \leq |E| < 1$, the position update is as follows:

$$X(t+1) = \Delta X(t) + E \cdot |J \cdot X_{rabbit}(t) - X(t)| \quad (6)$$

where J is characterized as a stochastic variable constrained within the interval $[0, 2]$; whereas $\Delta X(t) = X_{rabbit}(t) - X(t)$ represents the spatial difference between the prey and the current position of the individual.

2. When $r \geq 0.5$ and $|E| < 0.5$, the present status of the position is outlined as follows:

$$X(t+1) = X_{rabbit}(t) - E|\Delta X(t)| \quad (7)$$

3. When $r < 0.5$ and $0.5 \leq |E| < 1$, the present status of the position is outlined as follows:

$$X(t+1) = \begin{cases} Y, f(Y) < f(X(t)) \\ Z, f(Z) < f(X(t)) \end{cases} \quad (8)$$

Where $f(\cdot)$ represent the objective function; while the expressions for Y and Z are as follows:

$$\begin{cases} Y = X_{rabbit}(t) - E|J \cdot X_{rabbit}(t) - X(t)| \\ Z = Y + S \times LF(DIM) \end{cases} \quad (9)$$

Where J represents a stochastic variable constrained within the interval $[0, 2]$; while $\Delta X = X_{rabbit}(t) - X(t)$ denotes the spatial distance separating the location of the prey from the individual's current position. DIM refers to the dimensionality of the problem, S is a random vector characterized by DIM dimensions, and $LF(\cdot)$ signifies the Levy flight function.

4. When $r < 0.5$ and $|E| < 0.5$, the present status of the position is outlined as follows:

$$X(t+1) = \begin{cases} X_{rabbit}(t) - E|J \cdot X_{rabbit}(t) - X_m(t)|, & f(Y) < f(X(t)) \\ Z, & f(Z) < f(X(t)) \end{cases} \quad (10)$$

The HHO algorithm implements various position update strategies based on the aforementioned random states and subsequently assesses the fitness values for all individuals within the present generation. If an individual's fitness value surpasses that of the prey, the prey is substituted with this superior individual. The process

then continues into the next generation of iterations until the specified termination criteria are fulfilled.

Figure 7 illustrates the flowchart of the HHO algorithm.

Fitness function design

The parameters of the controller are critical to the performance of the control system. To this end, this paper employs the HHO algorithm to optimize the controller parameters. Meanwhile, the distance between adjacent points on the Poincaré section is selected as the criterion to construct the fitness function, as shown in equation (11), which effectively guides the HHO algorithm in searching for the optimal controller parameters (the global optimal solution that minimizes or relatively minimizes the fitness function corresponds to the optimal controller parameters). Additionally, this approach enables a quantitative evaluation of control performance (the relative minimum of the fitness function indicates the relatively optimal control effect).

$$\begin{aligned} f(\hat{P}_i) = & |d^* - \|X(k) - X(k-1)\|| \cdot \ln\left(\frac{1}{\eta}\right) \\ & + |0 - (1 + \text{sign}(r - 0.5))| \\ & + |\|X(k) - X(k-1)\| - \|X(k-1) - X(k-2)\|| \end{aligned} \quad (11)$$

where d^* is the expected value of the distance between adjacent points on the Poincaré section, $X(k)$ is the value of the state variable X , and η is a random number uniformly distributed in the interval $(0, 1)$, r is a random number.

The first term of equation (11; i.e. $|d^* - \|X(k) - X(k-1)\|| \cdot \ln\left(\frac{1}{\eta}\right)$), quantifies the progressive concentration of the scattered point set on the Poincaré section into a finite collection of discrete points, thereby embodying the transition from chaotic to desired periodic motion.

The second term (i.e. $|0 - (1 + \text{sign}(r - 0.5))|$) serves to adaptively adjust the controller's search behavior for the target periodic orbit: during the early iteration phase (determined by the proportional parameter ψ , where $\psi \in [0, 40\%]$ and increases with iteration steps), is set to maintain a higher fitness value, enabling the controller to explore a broader search space for the target periodic orbit. In the later iteration phase, is adjusted to drive the fitness value toward its minimum more rapidly, thereby enhancing the controller's convergence speed.

The third term (i.e. $|\|X(k) - X(k-1)\| - \|X(k-1) - X(k-2)\||$) quantifies the distance variation between adjacent points on the Poincaré section (the objective is to minimize this distance, ensuring that the points coalesce into a finite set, which signifies the successful control of chaotic motion into periodic motion).

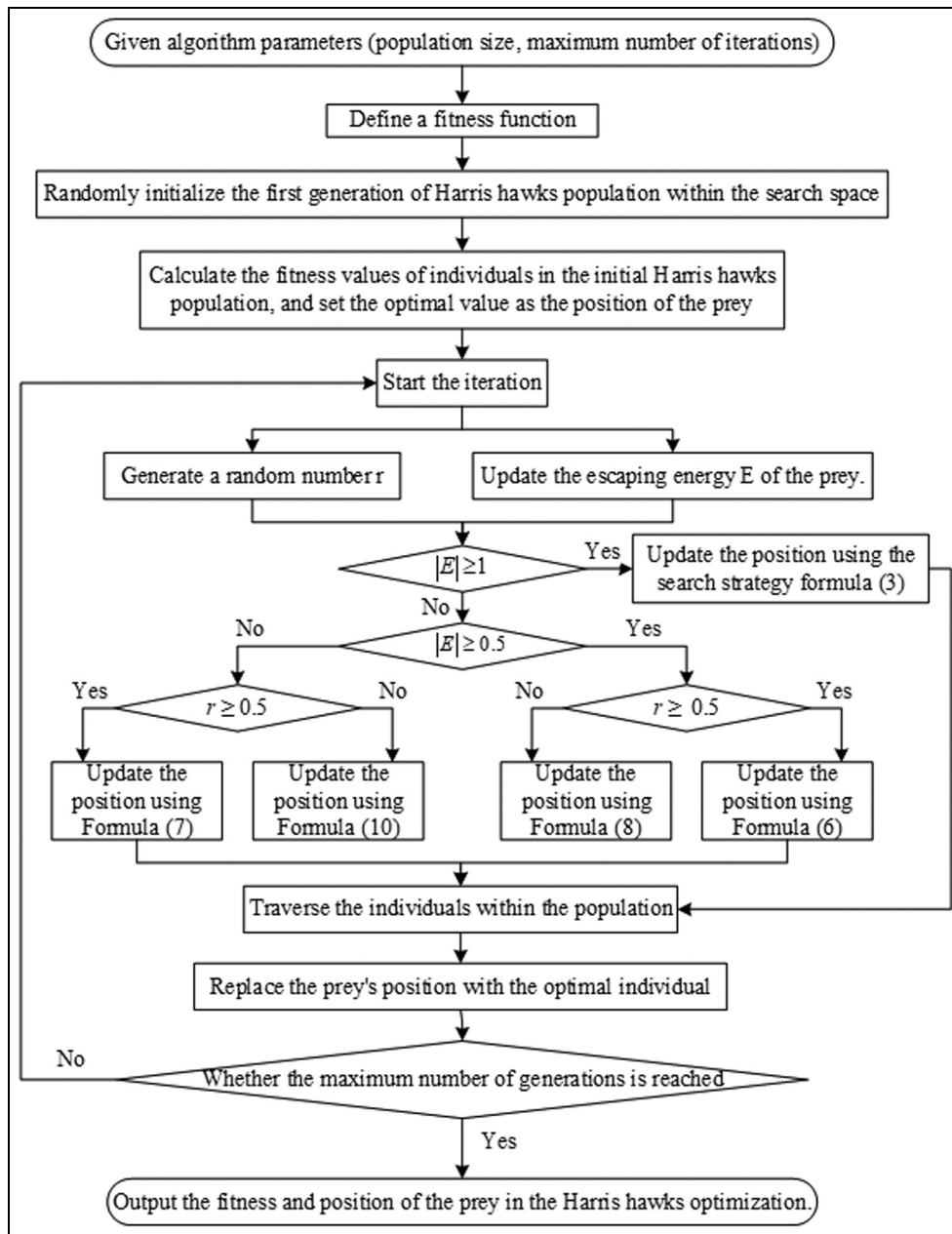


Figure 7. Flow chart of the HHO algorithm.

The designed fitness function effectively guides HHO to search for relatively optimal controller parameters. Combined with the controller's structural advantages, it can better learn the dynamic characteristic changes of the controlled system in input-output data, thus achieving better control quality.

Simulation and result analysis of chaotic control

HHO-DRBFNN simulation

To assess the feasibility and efficacy of the HHO-DRBFNN chaotic control approach, simulations are performed on the chaotic attractor depicted in Figure 4(a). Simulation conditions are set as follows:

HHO population size is 30; initial time step is 0.001 s, corresponding to a sampling rate of 1000 Hz; the system runs for 1000 steps, with the first 500 steps for observing uncontrolled chaotic motion and control applied at the 500th iteration to clearly demonstrate the chaotic control effect.

DRBFNN structure parameters: three input layer nodes, 3 first hidden layer nodes, 2 second hidden layer nodes (all using Gaussian activation functions with center range $[-1, 1]$), one output layer node with weight range $[-2, 2]$.

Controlled system parameters: damping coefficient $\zeta = 0.2$, restitution coefficient $R = 0.8$, clearance $h = 0.05$, excitation frequency in chaotic state $\omega = 2.655$ (the system is in chaotic state at this time).

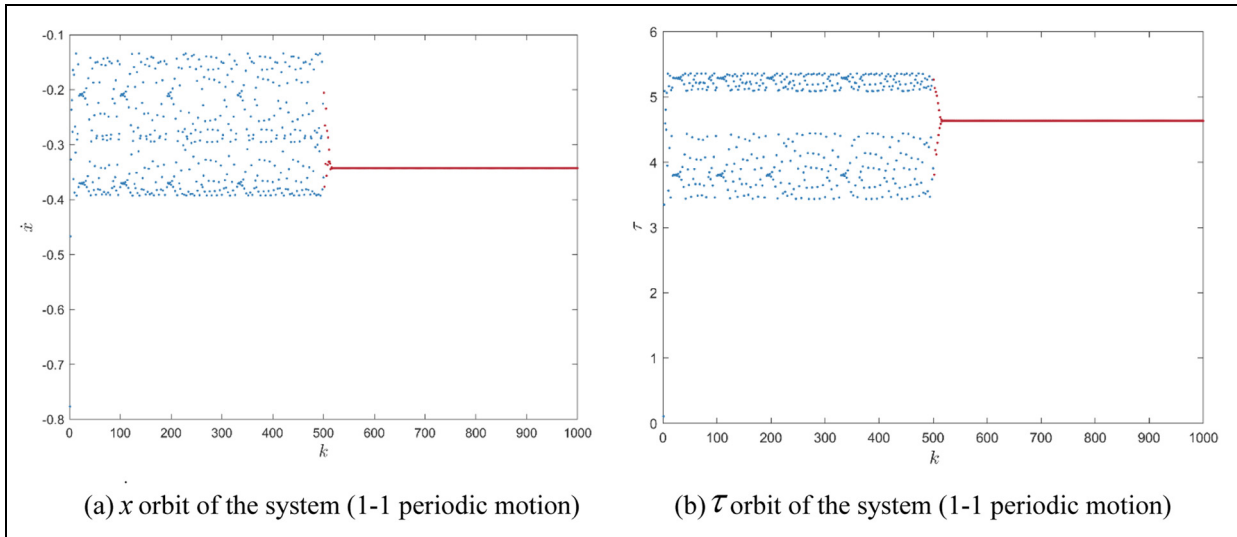


Figure 8. Regulated oscillatory behavior of the system (HHO-DRBFNN; 1-1 periodic motion): (a) x orbit of the system (1-1 periodic motion) and (b) τ orbit of the system (1-1 periodic motion).

Table 1. Parameters of controller (period 1-1 motion).

C_{sk}	σ_{sk}	W_k
(0.5935, -0.3371, -0.9135)	0.6207	-1.2768
(0.1811, -0.3960, -0.9215)	0.5847	-1.2611
(0.9158, -0.3264, -0.2225)	0.4293	-1.8280
(0.9861, 0.2441, -0.0147)	0.9057	-0.0055
(0.2370, -0.1067, -0.0513)	0.9087	1.0459

Table 2. Parameters of controller (period 2-2 motion).

C_{sk}	σ_{sk}	W_k
(0.8410, -0.3727, 0.4098)	0.3826	0.9261
(0.8334, 0.5904, -0.4425)	0.8284	-0.6521
(-0.5447, 0.5567, -0.4507)	0.2245	-1.0541
(-0.1025, -0.9329, -0.3728)	0.5296	1.9864
(0.6013, 0.3865, -0.6136)	0.6566	-0.0237

The absolute value of the control quantity u should be less than 0.015 to avoid system instability caused by excessive adjustment. The expected control targets are set to 1-1 periodic motion and 2-2 periodic motion. Due to space limitations, the control results of other periodic orbits are not shown.

Figures 8–10 show the orbit diagrams and phase diagrams of controlled periodic motion using the HHO-DRBFNN method. Tables 1 and 2 present the controller parameters that have been optimized using the HHO algorithm.

It can be seen from Figure 8 that chaotic motion is controlled at step 535 and stabilizes into 1-1 periodic motion. It can be seen from Figure 9 that chaotic motion is controlled at step 528 and stabilizes into 2-2 periodic motion. The closed curve in Figure 10(a) indicates that the system motion is currently stable 1-1 periodic motion. The phase diagram in Figure 10(b) shows

two closed curves. Therefore, the HHO-DRBFNN approach is capable of rapidly attaining efficient control over chaotic dynamics.

Comparative simulation analysis

HHO-RBFNN. HHO algorithm parameters: Same as HHO-DRBFNN (population size 30, maximum iterations 1000).

RBFNN structure parameters: two input layer nodes, five single hidden layer nodes, output layer weight range $[-2, 2]$.

Figures 11 and 12 show the orbit diagrams of controlled periodic motion using the HHO-RBFNN method. Tables 3 and 4 present the controller parameters that have been optimized using the HHO algorithm.

Due to space limitations, phase diagrams are not discussed. As shown in Figure 11, chaotic motion is controlled at step 581 and stabilizes into 1-1 periodic motion; as shown in Figure 12, chaotic motion is controlled at step 618 and stabilizes into 2-2 periodic motion.

The simulation results demonstrate that the HHO-RBFNN control approach successfully attains the desired control objectives.

Figure 13 shows the fitness curves of the HHO-RBFNN and HHO-DRBFNN control methods. It can be seen that HHO-DRBFNN has stronger continuous optimization ability and higher optimization accuracy. From the perspective of controller structures, HHO-RBFNN needs to increase more nodes to improve non-linear mapping ability, thereby enhancing its continuous optimization ability to obtain higher optimization accuracy; in contrast, HHO-DRBFNN reduces redundant nodes and connections through hierarchical feature extraction, and still has stronger continuous

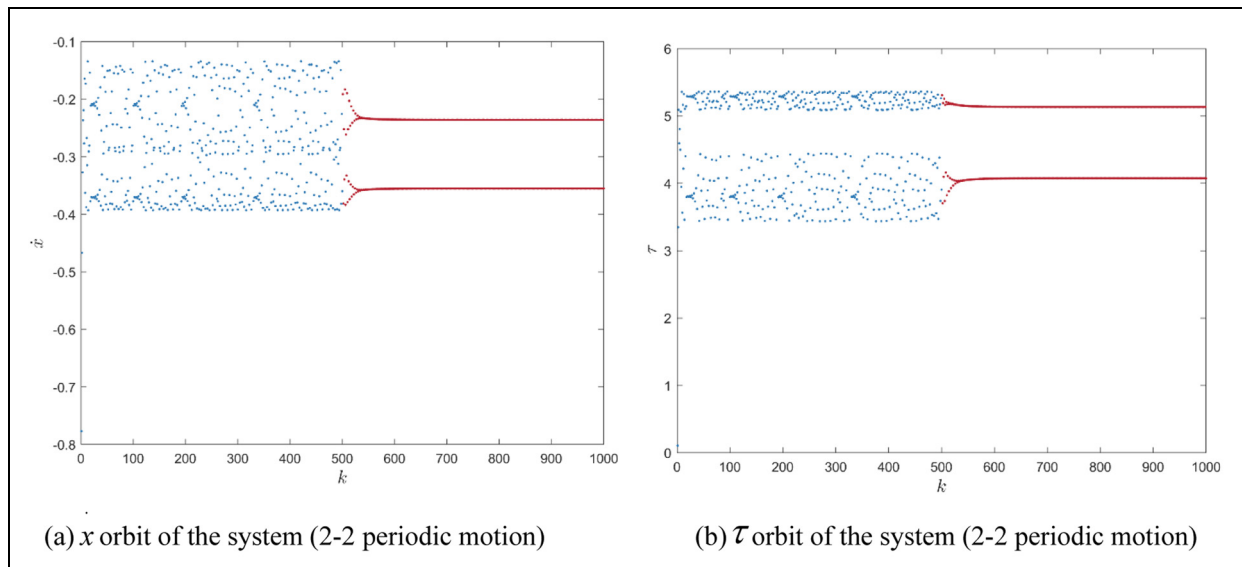


Figure 9. Regulated oscillatory behavior of the system (HHO-DRBFNN; 2-2 periodic motion): (a) x orbit of the system (2-2 periodic motion) and (b) τ orbit of the system (2-2 periodic motion).

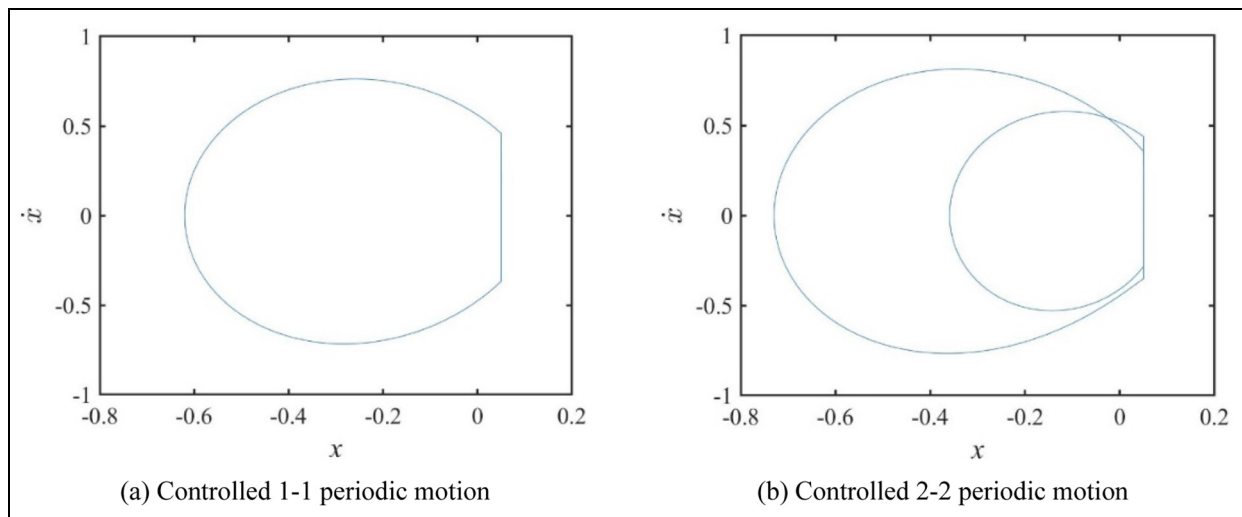


Figure 10. Phase diagrams of controlled periodic motion (HHO-DRBFNN): (a) controlled 1-1 periodic motion and (b) controlled 2-2 periodic motion.

Table 3. RBFNN controller parameters optimized by HHO (1-1 periodic motion).

C_{sk}	σ_{sk}	W_k
(0.3292, 0.2968, -0.0719)	0.9286	-0.4380
(-0.3924, -7055, -0.4588)	0.2728	-1.7248
(0.4050, 0.9008, -0.6610)	0.7762	-0.0141
(-0.6210, 0.7625, -0.4000)	0.5908	1.6005
(-0.6578, 0.6470, 0.8587)	0.3803	0.3727

optimization ability with relatively fewer nodes and structural parameters, thus obtaining higher optimization accuracy and improving parameter optimization efficiency.

TSA-DRBFNN. Tunicate Swarm Algorithm (TSA) algorithm parameters: Population size 30, maximum iterations 1000, exploration factor 0.6, exploitation factor 0.4.

DRBFNN structure parameters: Same as those in HHO-DRBFNN.

Figures 14 and 15 show the orbit diagrams of controlled periodic motion using the TSA-DRBFNN method. Tables 5 and 6 present the controller parameters optimized using the TSA algorithm. Due to space limitations, phase diagrams are not discussed.

As shown in Figure 14, chaotic motion is controlled at step 560 and stabilizes into 1-1 periodic motion; as shown in Figure 15, chaotic motion is controlled at step 607 and stabilizes into 2-2 periodic motion.

The simulation results demonstrate that the TSA-DRBFNN control approach successfully attains the desired control objectives.

Figure 16 shows the fitness curves of the HHO-DRBFNN and TSA-DRBFNN control methods. It

can be seen that HHO-DRBFNN has a faster convergence speed and smaller stable fluctuations; TSA-DRBFNN relies on manually set exploration and exploitation factors, which limits its convergence speed and accuracy.

Table 4. RBFNN controller parameters optimized by HHO (2-2 periodic motion).

C_{sk}	σ_{sk}	W_k
(0.4232, -0.2811, 0.0743)	0.7120	0.2785
(0.0171, 0.0269, 0.6986)	0.0367	0.9976
(0.7528, 0.5734, -0.6772)	0.3188	0.0864
(-0.4307, -0.4210, -0.2051)	0.4352	-1.2601
(0.2711, 0.0363, -0.6677)	0.8997	-1.2675

Comparative analysis of system simulation results. Through the above experimental comparison, it can be seen from Figures 8 and 9 that HHO-DRBFNN can stabilize the system from chaotic motion to 1-1 periodic motion and 2-2 periodic motion within 35 and 28 steps, respectively; it can be seen from Figures 11 and 12 that HHO-RBFNN can achieve stabilization within 60 and 107 steps; it can be seen from Figures 14 and 15 that TSA-DRBFNN can achieve stabilization within 81 and 118

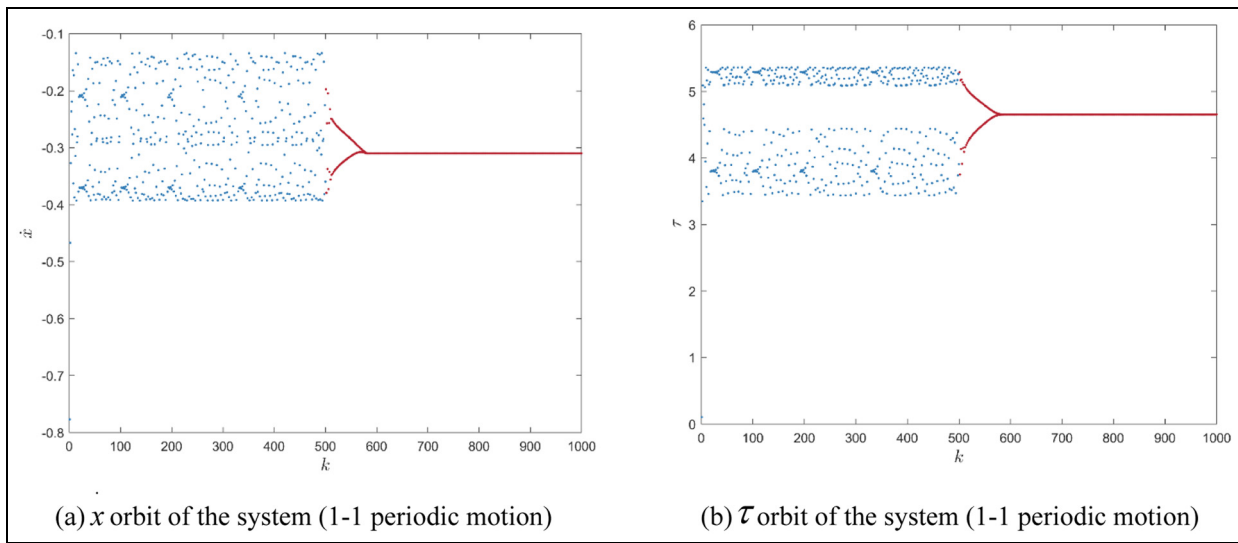


Figure 11. Regulated oscillatory behavior of the system (HHO-RBFNN; 1-1 periodic motion): (a) x orbit of the system (1-1 periodic motion) and (b) τ orbit of the system (1-1 periodic motion).

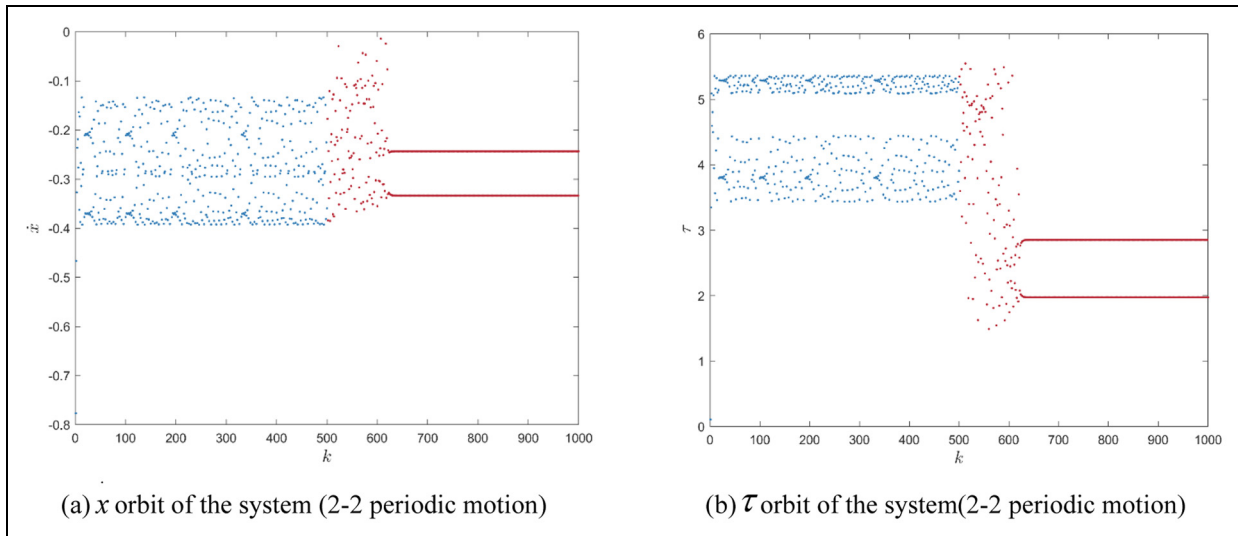


Figure 12. Regulated oscillatory behavior of the system (HHO-RBFNN; 2-2 periodic motion): (a) x orbit of the system (2-2 periodic motion) and (b) τ orbit of the system (2-2 periodic motion).

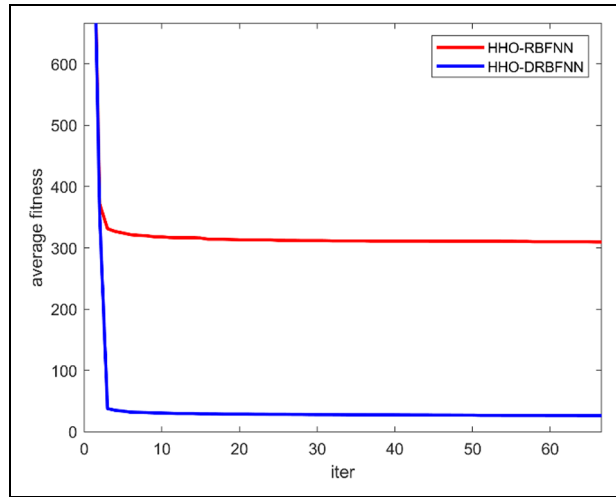


Figure 13. Fitness curves.

Table 5. Parameters of controller (period 1-1 motion).

C_{sk}	σ_{sk}	W_k
(0.1339, -0.0085)	0.0202	-0.0897
(0.0529, -0.0517)	0.0591	
(0.1476, -0.10333)	0.0570	0.0573
(0.0547, 0.0408)	0.1046	

Table 6. Parameters of controller (period 2-2 motion).

C_{sk}	σ_{sk}	W_k
(-0.0841, -0.2155)	0.1344	0.1793
(-0.2540, 0.2922)	0.1430	
(0.0026, 0.2413)	0.0906	0.0032
(-0.1073, 0.1310)	0.1833	

steps. Obviously, HHO-DRBFNN has a relatively faster convergence speed in terms of convergence time.

Based on the root mean square error (RMSE) data of the three control methods recorded in 30 experiments (as shown in Table 7), the average result of HHO-DRBFNN is 3.6784, that of TSA-DRBFNN is 4.0138, and that of HHO-RBFNN is 4.0623, indicating that the HHO-DRBFNN method can maintain high control accuracy in multiple tests and has smaller fitting errors for the strong nonlinearity and non-smoothness of the system.

Based on the standard deviation (STD) data of the three control methods recorded in 30 experiments (as shown in Table 8), the average result of HHO-DRBFNN is 3.3872, that of TSA-DRBFNN is 3.5246, and that of HHO-RBFNN is 3.6982. The STD of HHO-DRBFNN is significantly lower than that of HHO-RBFNN and TSA-DRBFNN, indicating that it is less sensitive to initial parameter perturbations and system uncertainties.

In summary, through the synergistic design of controller structure optimization, algorithm optimization mechanism, and fitness function, HHO-DRBFNN shows comprehensive advantages in control accuracy, robustness, and dynamic response speed. Multiple test results verify its reliability and stability in application to strongly nonlinear and non-smooth systems.

Discussion of simulation results

Simulation results show that in the chaotic control of single-degree-of-freedom vibro-impact systems, the proposed HHO-DRBFNN control method exhibits significant advantages in convergence speed, control accuracy, and dynamic stability compared with TSA-DRBFNN and HHO-RBFNN methods.

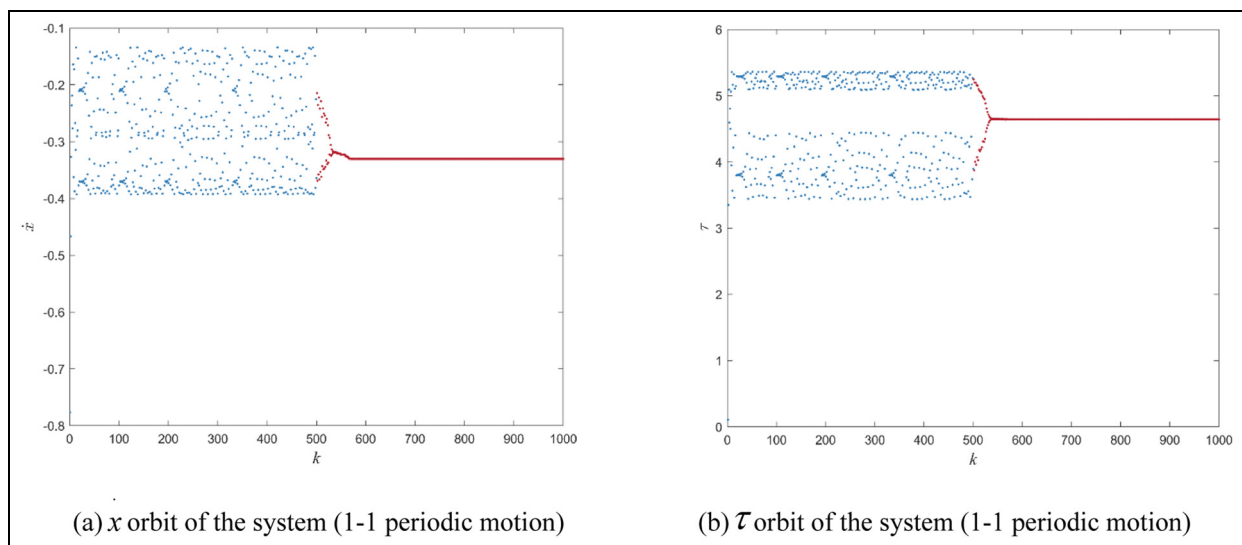


Figure 14. Regulated oscillatory motion of the system (TSA-DRBFNN; 1-1 periodic motion): (a) x orbit of the system (1-1 periodic motion) and (b) τ orbit of the system (1-1 periodic motion).

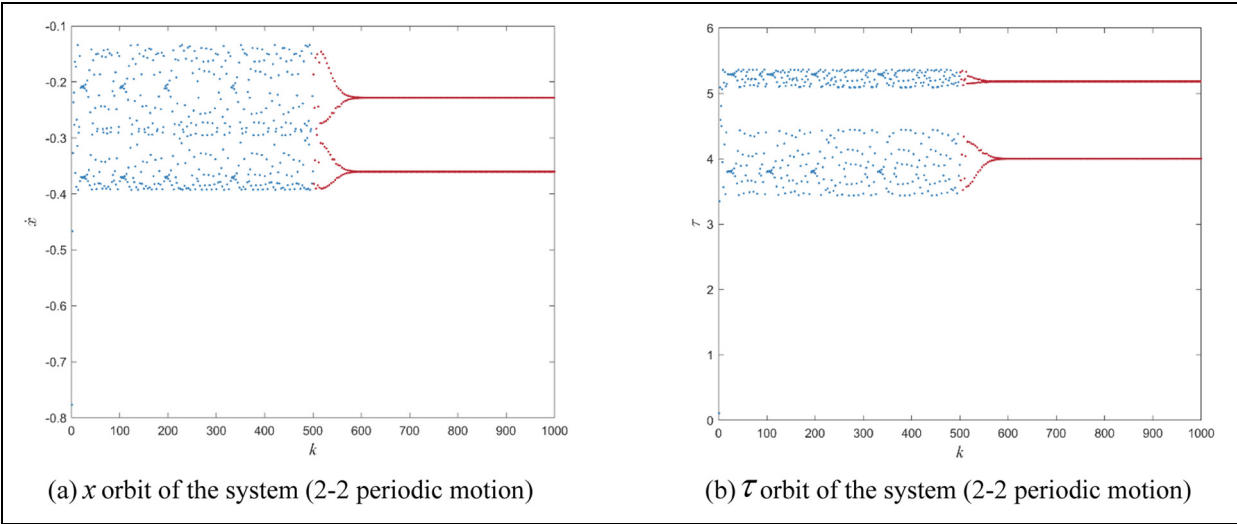


Figure 15. Regulated oscillatory motion of the system (TSA-DRBFNN; 2-2 periodic motion): (a) $\dot{x}$ orbit of the system (2-2 periodic motion) and (b) τ orbit of the system (2-2 periodic motion).

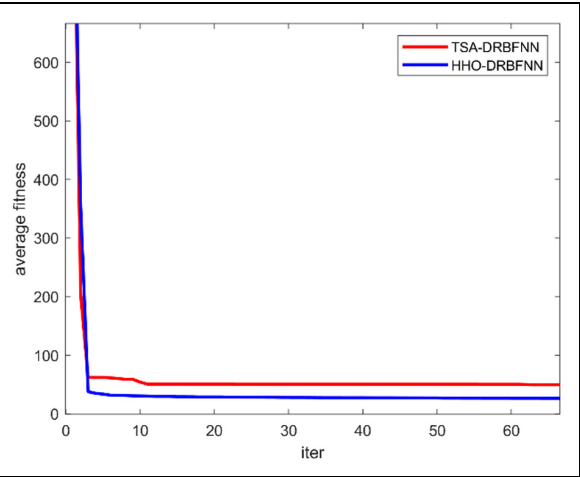


Figure 16. Fitness curves.

This performance difference can be systematically analyzed from three dimensions: controller structure design, optimization algorithm mechanism, and rationality of the fitness function, as follows:

1. Mechanism of enhanced nonlinear fitting ability by DRBFNN dual-hidden-layer structure

From the perspective of structural design, the DRBFNN controller reduces parameter redundancy through hierarchical feature extraction, enhancing the ability to learn system nonlinear characteristics.

2. Driving effect of HHO’s adaptive optimization mechanism on parameter optimization efficiency

The optimization mechanism of HHO ensures efficient parameter optimization of DRBFNN, significantly improving parameter search efficiency and

Table 7. Root mean square error (RMSE) of three control methods.

Test no.	HHO-DRBFNN	TSA-DRBFNN	HHO-RBFNN
1	0.2686	4.7310	0.9276
2	0.2731	1.6421	6.3673
3	0.4705	5.0499	1.9610
4	5.1985	2.0161	2.9986
5	3.0619	11.522	3.1930
6	1.9277	0.4633	0.0729
7	9.4391	6.1505	11.5405
8	4.0744	8.4057	5.2271
9	2.8261	2.8435	6.3343
10	3.4800	5.9382	0.7665
11	8.5277	1.1002	1.0881
12	4.5732	5.9197	2.2378
13	0.7679	2.8151	7.2250
14	2.3048	1.1840	0.5650
15	3.4250	8.6863	0.7678
16	4.2305	0.6117	4.7565
17	0.8721	0.0640	8.6486
18	3.4509	0.8773	1.3404
19	1.3889	9.7123	4.1028
20	2.2496	6.3347	3.5772
21	1.4262	1.1217	0.2555
22	1.9480	9.9284	2.4659
23	9.0294	0.5102	2.8703
24	7.0133	1.5764	6.3692
25	1.4538	12.1495	3.7484
26	0.1884	5.43056	5.9205
27	6.9789	1.5611	10.0000
28	2.1210	4.4895	4.2774
29	11.3019	1.3914	8.5980
30	0.0768	0.7943	1.0140

accuracy compared with TSA-DRBFNN and HHO-RBFNN.

3. Directional guidance of the fitness function based on Poincaré section features on optimization direction

Table 8. Standard deviation (STD) of three control methods.

Test no.	HHO-DRBFNN	TSA-DRBFNN	HHO-RBFNN
1	0.2442	4.2990	0.8648
2	0.2257	1.4752	5.8234
3	0.4077	4.5780	1.7644
4	4.7041	1.8154	2.7287
5	2.7837	10.4581	2.8980
6	1.7321	0.4218	0.0516
7	8.5097	5.5572	10.5146
8	3.6918	7.6082	4.7789
9	2.5630	2.5869	5.7667
10	3.1530	5.3690	0.6857
11	7.6941	0.9823	0.9870
12	4.1123	5.3650	2.0553
13	0.6846	2.5622	6.5867
14	2.0876	1.0600	0.4997
15	3.1017	7.8637	0.7055
16	3.8110	0.5583	4.3507
17	0.7767	0.01998	7.8991
18	3.1223	0.7741	1.2239
19	1.2558	8.7985	3.7469
20	2.0246	5.7455	3.2496
21	1.2941	1.0034	0.2121
22	1.7689	8.9844	2.2555
23	8.1700	0.4710	2.6182
24	6.3422	1.4311	5.8137
25	1.3186	11.0227	3.4371
26	0.1690	4.9394	5.3915
27	6.2948	1.3953	9.1070
28	1.9224	4.0784	3.9061
29	10.2112	1.2547	7.8250
30	0.0560	0.7065	0.9105

The fitness function based on the distance between adjacent points on the Poincaré section directly correlates with the essential difference between chaotic and periodic motions. Through qualitative description and quantitative characterization of Poincaré sections and chaotic/periodic motions, it achieves more accurate parameter optimization than traditional error indicators.

Conclusions

A parameter feedback chaos control approach utilizing HHO-DRBFNN is introduced to address the challenge of controlling chaotic motion in a specific category of vibration impact systems characterized by clearance. This method enhances the efficiency of chaos control and offers a theoretical foundation and methodology for practical implementation.

- (1) The dynamics of the system under harmonic excitation frequency have been elucidated, leading to the establishment of criteria for the analysis of bifurcation and chaos parameters.
- (2) A chaos controller utilizing a DRBFNN has been developed. This controller features a dual hidden layer architecture, which, in contrast to conventional RBFNN, requires fewer

parameters for optimization, exhibits enhanced nonlinear fitting capabilities, and demonstrates superior adaptability to the pronounced nonlinear and nonsmooth characteristics inherent in the system. In order to optimize the parameters of the controller, an advanced HHO algorithm is utilized. The fitness function is constructed based on the distance between two consecutive points on the Poincaré section, which aids in the identification of optimal parameter solutions and subsequently improves the performance of the controller.

- (3) With the goal of controlling chaotic motion into 1-1 periodic motion and 2-2 periodic motion, simulations are conducted for TSA-DRBFNN, HHO-RBFNN, and HHO-DRBFNN control methods. Results show that all three methods can achieve the expected control target, but the HHO-DRBFNN control method has obvious advantages, with faster convergence speed, higher accuracy, better ability to learn system nonlinear characteristics, smaller fitness values in the same number of iterations, and more stable fitness curves in the later search stage.

Despite the good chaotic control performance of the HHO-DRBFNN method in single-degree-of-freedom impact vibration systems, there are still limitations:

- (1) The scalability of the approach to systems with multiple degrees of freedom requires additional investigation: The application of neural networks to the chaotic control of multi-degree-of-freedom vibro-impact systems has not been documented in the existing scholarly literature. The research team is conducting relevant exploration but requires further in-depth study.
- (2) High sensitivity to training data: The optimization results of HHO are easily affected by the distribution of initial training samples, and sample bias may reduce the adaptability of controller parameters.
- (3) Complexity in parameter tuning: HHO performance is affected by hyperparameters, requiring multiple tests to increase debugging costs; DRBFNN hidden layer nodes need manual adjustment, and adaptive mechanism exploration is needed in future research.

In the domain of chaos control, it is essential to further advance the integration and innovation of sophisticated algorithms, including the HHO algorithm, alongside diverse control strategies.

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Ethical considerations

This study does not involve human subjects, human tissues, or experimental animals. All data used in this research are derived from publicly available databases or de-identified secondary data that have been properly authorized for academic use. No procedures in this study require approval from an Institutional Review Board (IRB) or Animal Ethics Committee (AEC). All research activities strictly comply with the ethical guidelines of the Declaration of Helsinki and relevant international academic norms.

Consent to participate

Written informed consent for publication was obtained from all participants.

Author contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Ningzhou Li, Xinyu Bai, Xiaojuan Wei, Xiaoqi Li and Qingli Zhang. The design and execution of experiments, and the organization and analysis of experimental data were completed by Xinyu Bai. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

All the data are presented in the paper.

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